

Keeper K943 — FORCING-CHAIN AUDIT: is D_IV⁵ FORCED or FITTED?

VERDICT: CONDITIONAL. The forcing is real in architecture — the derivation runs observation → rank=2 → Type IV → n=5 → integers → physics, with physics as OUTPUT and the upstream premises minimality/robustness (NOT “match the SM”). That is the honest, defensible rebuttal to the hostile reviewer’s “you fitted a domain to known physics.” BUT it is more a forcing CHAIN than a fan of independent roads, and it has three asserted/conditional nodes that a hostile CI will target. Two of Casey’s specific phrasings need softening. Verdict feeds Lyra’s §3.

[PROGRAM: TEGMARK] Keeper, 2026-07-27. The #1 load-bearing claim for the Tegmark/MUH pitch. Based on a full corpus sweep (Paper52, grace_Cartan_elimination_sweep, T944, F571→F642→F647→K813, Cal referee 2026-07-15, Casey_Principle_16, HonestState ledger v0.11).

THE HONEST ANSWER (forced vs fitted)

FORCED, in architecture. The master derivation (BST_Paper52_Two_Five_Derivation) genuinely places physics DOWNSTREAM: the upstream premises are *observation exists*, *observation is structurally stable*, *depth economy* — minimality/robustness, not SM-matching. No “3 colors / 3 generations / dim 5” enters as an input. This is the real rebuttal to “fitted” and it is defensible.

But the strength is CHAIN-like, not over-determined. The arguments are largely SEQUENTIAL (rank → type → dimension), each with its own premise, not N independent roads that each suffice. Genuine redundancy exists at ONE node — n=5 is doubly-determined (genus coincidence AND short-root multiplicity a=3). Elsewhere it is single-threaded, so the chain is only as strong as its weakest link — and two links are asserted, not proved.

THE FORCING FAMILIES (what the corpus actually contains)

Family	Argument	Premise type	Tier
Rank=2	triangulation (≥ 2) + depth economy (≤ 2)	minimality	T944, framework/asserted
Type IV	rank stable under dimensional perturbation; I–III fail (rank $\propto$ dim), V isolated	robustness	Paper52 §4.2, asserted not proved
n=5 (route 1)	genus coincidence $n+2 = 2n-3$	self-consistency	rests on BST-constructed spectral genus
n=5 (route 2)	short-root multiplicity $a = n-2 = 3$ (Faraut–Korányi); $a=3$ only for D_IV ⁵	classification	SOLID as classification — but see Node 1
Chirality	n_C odd → Dolbeault index ± 4 (chiral split)	structural	THEOREM (domain) — orthogonal to observed parity
137	$N_C^3 \cdot n_C + \text{rank} = 137$, matches α^{-1} (0.0001%)	downstream property	match solid; cross-form exclusivity UNTESTED
Conformal FP	$SO(4,2) \subset SO(5,2)$, $\beta=0$ (scale-free)	structural property	confirms, doesn’t eliminate others

The classification step itself is SOLID: grace_Cartan_elimination_sweep does a real per-type walk (D_I, D_II, D_III, D_IV, E_III, E_VII) and only D_IV⁵ has (rank=2 $\wedge$ dim=5). That “unique GIVEN the criteria” is proper Cartan-classification math.

THE THREE NODES A HOSTILE CI WILL HIT

Node 1 — Criteria-selection (framework-asserted; + a smuggling risk). The classification proves “unique GIVEN $\text{rank}=2 \wedge \text{dim}=5$ ”; that these are THE necessary substrate requirements is a framework call (Grace flags this herself). Sharp sub-issue to resolve: the “non-circular” version swaps $\text{dim}=5$ for short-root multiplicity $a=3$ — but $a=3 = \text{three short roots} = N_c=3 = \text{three colors}$. If $a=3$ is “3 colors” in a geometry hat, the non-circularity moved the fit, it didn’t remove it. Must resolve whether $a=3$ is structural or physics (color-count).

Node 2 — The genus coincidence rests on a BST-constructed quantity. $n=5$ (route 1) sets $n+2 = 2n-3$, but “ $2n-3$ ” is a *spectral genus* the corpus DEFINES itself, not a standard invariant (standard arithmetic genus of Q^n is 0). Authors flag it in-source (Paper52 §5.1: “BST-specific computation, not a standard result”; falsifier P5 concedes fragility). Highest reverse-engineering-risk node. *Mitigation*: $n=5$ route 2 ($a=3$ multiplicity) is independent of the genus construction — so $n=5$ does NOT hinge solely on the constructed quantity (pending Node 1).

Node 3 — Chirality $\rightarrow$ observed parity was REFUTED, now conditional. The clean “ $g=7$ odd $\rightarrow$ weak force chiral” lock was refuted by the team’s OWN operator computation (F642: $\{\Sigma_{06}, \gamma^5\}=0$ — holomorphicity ∇ 4D spacetime parity). Domain chirality (Dolbeault ± 4 , n_C odd) survives as a THEOREM; observed 4D parity violation is now DERIVED-CONDITIONAL on a gravi-weak input F647 says is NOT computed. V-A via F(4) super-grading is CANDIDATE tier (and F(4) is an extension beyond D_{IV}^5). Not forced — open.

TWO THINGS BETTER THAN A SKEPTIC WOULD EXPECT

- Chirality is TARGET-INNOCENT. $g=7$ was fixed in the 2022 five-integer set, before any parity/weak work; Cal (referee) affirmed on record (2026-07-15) that the chirality argument uses ONLY $g=7$ ’s oddness, zero reference to observed parity. So it is NOT physics-imported — just an unfinished derivation.
- 137 is a downstream PROPERTY, not a fitted selector. Computed once $(N_c^3 \cdot n_C + \text{rank})$, then matches α^{-1} . It does not choose the domain — so it can’t be “the constant we tuned to.”

TWO OF CASEY’S PHRASINGS TO SOFTEN (Keeper honesty)

- “137 capacity found ONLY on D_{IV}^5 ” — the cross-form census was NEVER RUN. Grace filed it as a next step; nobody computed the capacity on $D_I/D_{II}/D_{III}/E_{III}/E_{VII}$. Plausible, but currently UNTESTED, not a result. $\rightarrow$ ACTION: run it (cheap, high-value).
- “other forms lack chirality” — right about *domain* chirality (oddness), overstates the link to *observed* parity given the F642 refutation.

ACTIONS (queued)

1. ★ Elie — the 137 cross-form census (toy). Compute $(\text{short-root-count}^3 \cdot \text{dim}_C + \text{rank})$ for all six irreducible Hermitian symmetric families; is 137 unique to D_{IV}^5 ? Confirms Casey’s phrasing + banks an independent selector, OR corrects it. Finite, reviewer-runnable.
2. ★ Keeper/Lyra — resolve Node 1’s sub-issue: is $a=3$ structural or $= N_c=3$ (color count)? Decides whether the “non-circular” criterion is genuinely physics-free.
3. ★ Lyra — §3 writes the ARCHITECTURE, tiered. Claim: constraints structural, physics downstream, upstream premises minimality-not-SM. Label the three nodes at their tier. Do NOT claim “137 only on D_{IV}^5 ” or “chirality forced by oddness” as clean locks.
4. Spectral genus $g=2n-3$ — flag for grounding-or-labeling (standard invariant? or honest BST-construction label). Research item.
5. Chirality F647 Route-2 boundary computation — the open node; already tracked; do not claim forced until computed.

Consistency notes

- Registry marks T953/T944 “Proved”; HonestState ledger v0.11 tiers the uniqueness “RATIFIED + CANDIDATE.” The ledger is the more honest current tier — flag the registry as over-stated for this entry (stamp

on reverse-walk).

- Three non-identical elimination logics coexist (Grace sweep: $\text{rank} \wedge \text{dim}/a$; Paper52: rank-stability; T944: genus). They agree on the answer but are not one canonical proof — a signature of assembled plausibility. A single canonical uniqueness theorem would strengthen the pitch.

ADDENDUM (second-pass sweep) — independence count + the leg to defend

The expanded sweep (all forcing lines enumerated) sharpens the verdict in three ways:

1. The over-determination framing was ALREADY RETRACTED by the team — internally. This is important and GOOD (internal honesty, not a live overclaim): - Lyra Strong-Uniqueness v1.6 explicitly withdraws the null-model: the 11 legs are “convergence-of-routes evidence, NOT independent multiplicative null-model confirmations.” The earlier $(1/3)^{11} \approx 5.6e-6$ claim is WITHDRAWN. - Grace (Cal #330): “these are NOT four independent proofs — criterion ($\text{rank}=2 \wedge \text{dim}=5$) alone is airtight; (1)–(3) are robustness.” - T944 “One-Input Theorem”: the five integers are five READINGS of one object; $\text{rank} = n_C - 3$ links them (specify either, the other follows). - Honest independent-structural-road count ≈ 2 ($\text{rank}=2$ from observation; $n_C=5$ from genus/Wallach), arguably 1 (the one-input collapse). DISCIPLINE: §3 and the hook must NOT resurrect “N independent forcings” — the team already killed that framing. Claim convergence-of-routes on ONE object, honestly.

2. The single leg worth DEFENDING: T1829 scalar-Wallach equality (BST_PaperB...Cartan_elimination_v0.6; toy 2151, 26/26; PROVED, dimension-free). $d_0=1 \Rightarrow N_c = \text{rank}^2 - 1 = 3$, forcing $n=5$. This is the one rigorous rep-theoretic road that is NOT a re-reading of the constructed spectral genus (Node 2). §3 should LEAD with T1829, not the genus coincidence — it’s the strongest proved single forcing and it sidesteps the reverse-engineering-vulnerable node.

3. Node 1’s $a=3$ -smuggling worry moves toward FORCED (pending verification). If $N_c = 3 = \text{rank}^2 - 1$ comes STRUCTURALLY from $\text{rank}=2$ via T1829’s scalar-Wallach equality, then $a=3$ ($= N_c = \text{short-root count}$) is forced BY THE RANK, not fitted to “three colors.” → ACTION: verify T1829’s $\text{rank}^2 - 1$ does not itself smuggle $N_c=3$ in a premise. If clean, Node 1’s sub-issue resolves in the forced direction (partial upgrage).

Gauge-hosting confirmed OUT of the spine (good). The $\text{SO}(5) \times \text{SO}(2) \rightarrow \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ mechanism is the one genuine SM-premise risk; the corpus flags it OPEN (multi-month candidate) and correctly does NOT use it to force D_{IV}^5 . One minor gauge nod remains (T953 “ $N_c \geq 3$ prime for asymptotic freedom”) but Paper52 re-derives $N_c=3$ as consequence, not premise.

Revised one-line state: *forced in architecture, thin in independent evidence — and the team already knows it’s thin (null-model retracted)*. The leg to defend is T1829 (proved); the seam to flag is the spectral-genus definition (Paper52 §5.1, author-flagged); the two physics-flavored contacts (observed chirality, 137-uniqueness) are respectively conditional and un-computed. Not SM-fitted at the premise; not a plausibility mush; but not the “ten roads to one domain” the early framing suggested.

CENSUS VERDICT (Cal §92 + Keeper independent verification) — Node 1 RESOLVED for the integer, two soft spots named

Cal ran T1829’s $N_c = \text{rank}^2 - 1$ across the Cartan classification; Keeper independently verified (multiplicity table, arithmetic, proof target-innocence, the $\text{rank}=2$ premise, E7). Result: the census is CORRECT and it sharpens Node 1 substantially — but leaves two honest soft spots, both now precisely isolated.

CONFIRMED (SOLID): - The characteristic multiplicity table $\{I, II, III, IV, E_{III}, E_{VII}\} \rightarrow a = \{2, 4, 1, n-2, 6, 8\}$ is STANDARD (matches Faraut-Korányi/Helgason; passes a hostile cross-check). Filed in *grace_Cartan_elimination_sweep* + Paper126. - $N_c = \text{rank}^2 - 1$ is satisfied by EXACTLY $\{D_{IV}^5$ ($a=3$, $\text{rank } 2$), E7 ($a=8$, $\text{rank } 3$) — arithmetic independently reproduced. Types I/II/III/ E_{III} never satisfy it for integer rank. - THE CRUX HOLDS (the key win): T1829’s proof is PHYSICS-FREE. $d_0 = \text{rank}^2 / (N_c + 1) = 1$ (scalar Hardy space, lowest K-type = the constants) → $N_c = \text{rank}^2 - 1 = 3$ from $\text{rank}=2$ alone. Pure representation theory, ZERO SM input. Node 1’s $a=3$ -smuggling worry resolves in the FORCED direction FOR THE INTEGER: the 3 comes from the geometry, not from “three colors.”

TWO SOFT SPOTS NAMED (both already known to the team; neither is a physics contaminant INSIDE T1829):

1. rank=2 is a PREMISE (structural but asserted), and it is DOUBLY load-bearing. It is structural-minimality (triangulation + rank-1 disk degeneracy, T944) — NOT observed-generation-count — but it is not eliminated, and it is the SOLE discriminator between D_{IV}^5 and its genuine co-solution E7. A hostile reviewer’s cleanest attack: “your census gives TWO domains; you pick D_{IV}^5 over E7 by an asserted premise.” Correct — state it, don’t hide it. (Note: Cal’s §92 “rank=2 = smallest supporting 3=rank+1 generations” is MORE physics-flavored than the corpus’s own structural argument; use the structural form.)
2. ****geometry-3 $\rightarrow$ SU(3)_color is a SEPARATE, corpus-admitted OPEN identification.**** The INTEGER 3 is forced; the COLOR GROUP is not. Grace’s own walk-back: at restricted-root level the group is SO(3), so “ $N_c=3=a$ ” is a dimension-match ($3=3$), not a color derivation. Paper B: “value-recurrence; color downstream, open (#418).” Claim “the integer 3 is forced, not fitted”; do NOT claim “QCD color is forced.”

TWO OVER-ATTRIBUTIONS §3 MUST AVOID: 1. “T1829-proved relation across all six families” — toy 2151 proves it for TYPE IV ONLY ($n=3,5,7,9$ scan); the E7 hit + cross-family census is a fresh hand computation, correct but NOT toy-verified. Say “T1829 proves it for type IV; the classification census extends the arithmetic to all six families.” ($\rightarrow$ this is Elie’s task #28 to toy-verify.) 2. “ $N_c=3$ = three colors” — see soft spot 2.

Node 1 status: RESOLVED for the integer (forced, physics-free); residual = the rank=2 premise (doubly load-bearing, incl. E7) + the open color-group identification. Task #29 DONE. The E7 gap is the target of the third prong (see below).

THE THIRD PRONG (Casey, 2026-07-27) — inverse/rigidity, and it closes the E7 hole

Casey proposed a THIRD KIND of forcing: not another forward road (those were retracted) but the INVERSE direction — outcome $\rightarrow$ manifold (“the observed physics constrains the initial manifold”). Forward=existence, inverse=identifiability; both together = well-posedness (a mathematical property, obstinacy-resistant). Validity gate (4 conditions): full candidate set; UNIFORM functor (each domain uses ITS OWN integers, no D_{IV}^5 machinery leaking); DATA selector (measured, not BST-derived); don’t count reproducing the integers (=fitting).

Concrete payoff — the inverse route CLOSES the E7 gap that the structural census leaves to an asserted premise. E7 (rank 3) has rank+1 = 4 boundary strata $\rightarrow$ predicts 4 generations (Korányi-Wolf, a uniform fact). We OBSERVE 3. So observed-3-generations is DATA that forces rank=2 and EXCLUDES E7 — without invoking the asserted structural minimality premise. The structural route picks D_{IV}^5 by asserting rank=2; the inverse/data route picks it because E7 predicts the wrong generation count. The third prong does real work on the one honest hole. (Gate check needed: confirm “generations = rank+1 strata” is the general Korányi-Wolf identification applied uniformly, not a D_{IV}^5 -specific relation — condition 2.)

Build: the rigidity-under-data toy — give each of the ~6 nearest candidate domains ($D_{IV}^4, D_{IV}^6, D_{III}^2, E_{III}, E7, \dots$) its OWN integers, compute the same clean SM observables (generation count, $\alpha^{-1}=N_c^3 \cdot n_C + \text{rank}$, color factor, ...) from those integers, show the measured SM lands only on D_{IV}^5 and every neighbor misses by many σ . Keeper pre-registers the “miss” thresholds BLIND; Elie builds (extends task #28). CMB/ n_s = a candidate FOURTH prong, audited separately (n_s depends on g,rank; I/S-tier, own audit before any claim).

— Keeper K943, 2026-07-27 [TEGMARK]. CONDITIONAL: forced in architecture (physics downstream, minimality upstream — the real rebuttal); three asserted/conditional nodes (criteria-selection + $a=3$ -smuggling-risk; BST-constructed spectral genus; refuted-then-conditional chirality); two better-than-expected (target-innocent chirality; 137 = downstream property); two Casey phrasings softened (137-cross-form UNTESTED; chirality-of-other-forms overstated). Companion: [[BST_hook_paper_OPENING_SPEC_for_Lyra_2026-07-26]], [[BST_What_We_Claim_And_Do_Not_scope_page_for_hook_package_2026-07-26]].